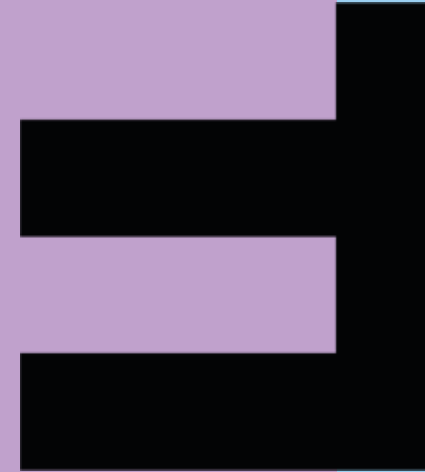


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Technical Information Systems

Entity Relationship Modelling – Exercises 2

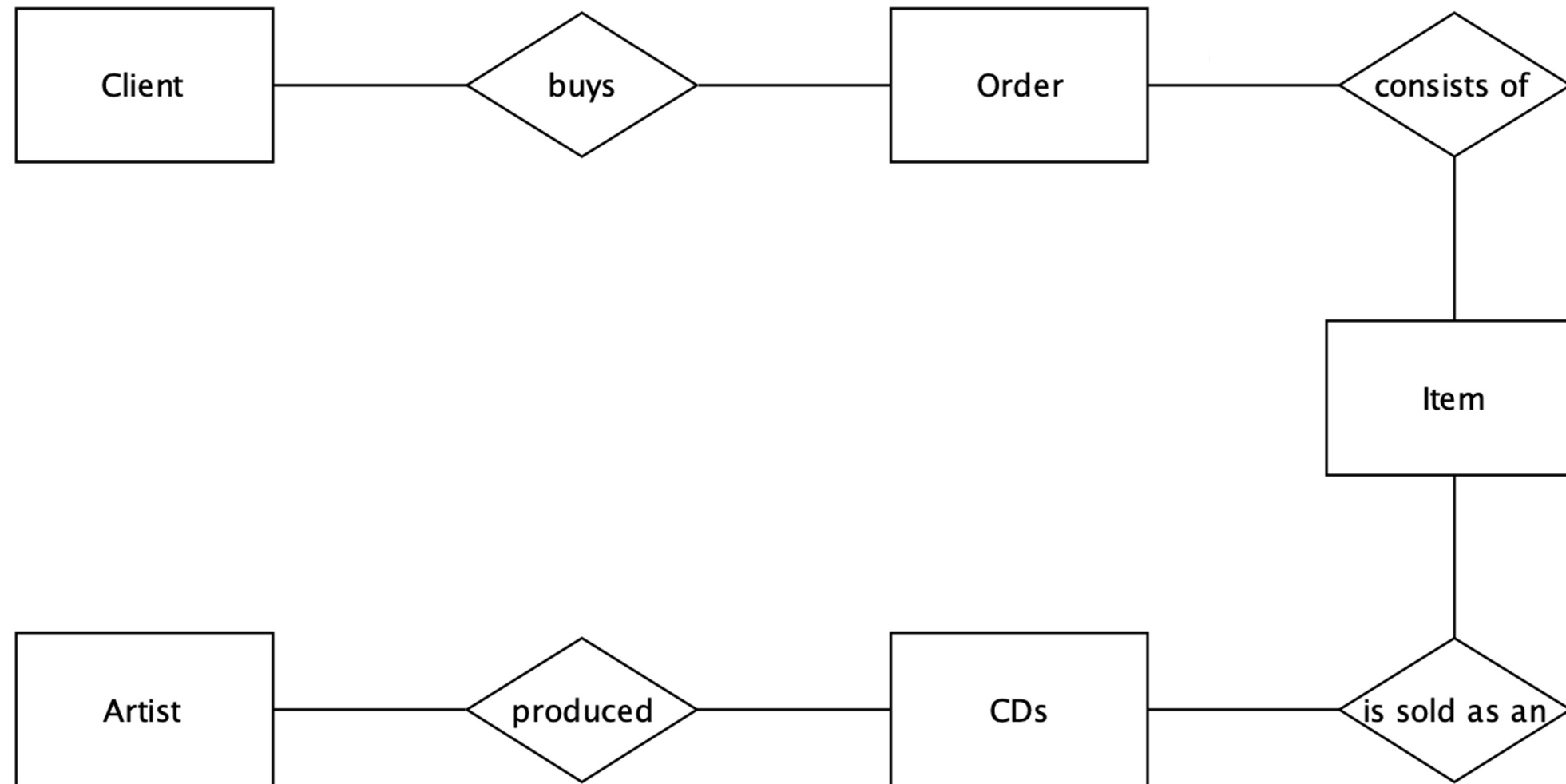
Music shop (simplified)

- You create a database for an online music shop
 - Each client can buy several CDs in one order
 - Each CD was recorded from one artist only
 - Artists might have produced more than one CD
- Create an ER model (without attributes and in a first step without cardinalities and optionalities, in a second step add them)
- Any questions?
-
- We want the addresses of potential clients (before their first purchase)
 - An artist must at least have one CD to be in our database
-
- The price of the CDs does not change
 - Each CD can be identified by the International Article Number (IAN)

Music shop – Chen Diagram

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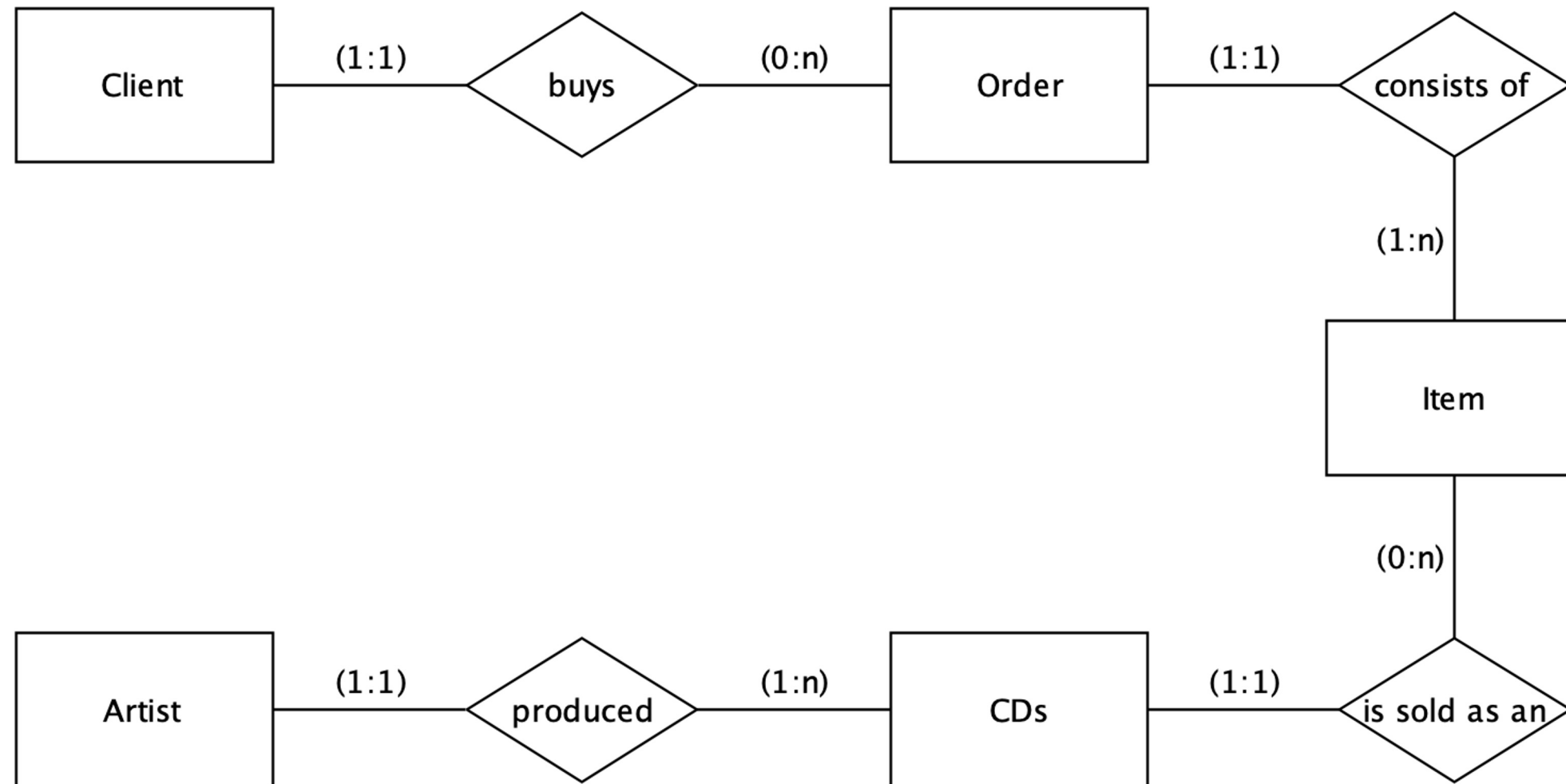


- Now add cardinalities and optionalities

Music shop – Chen Diagram

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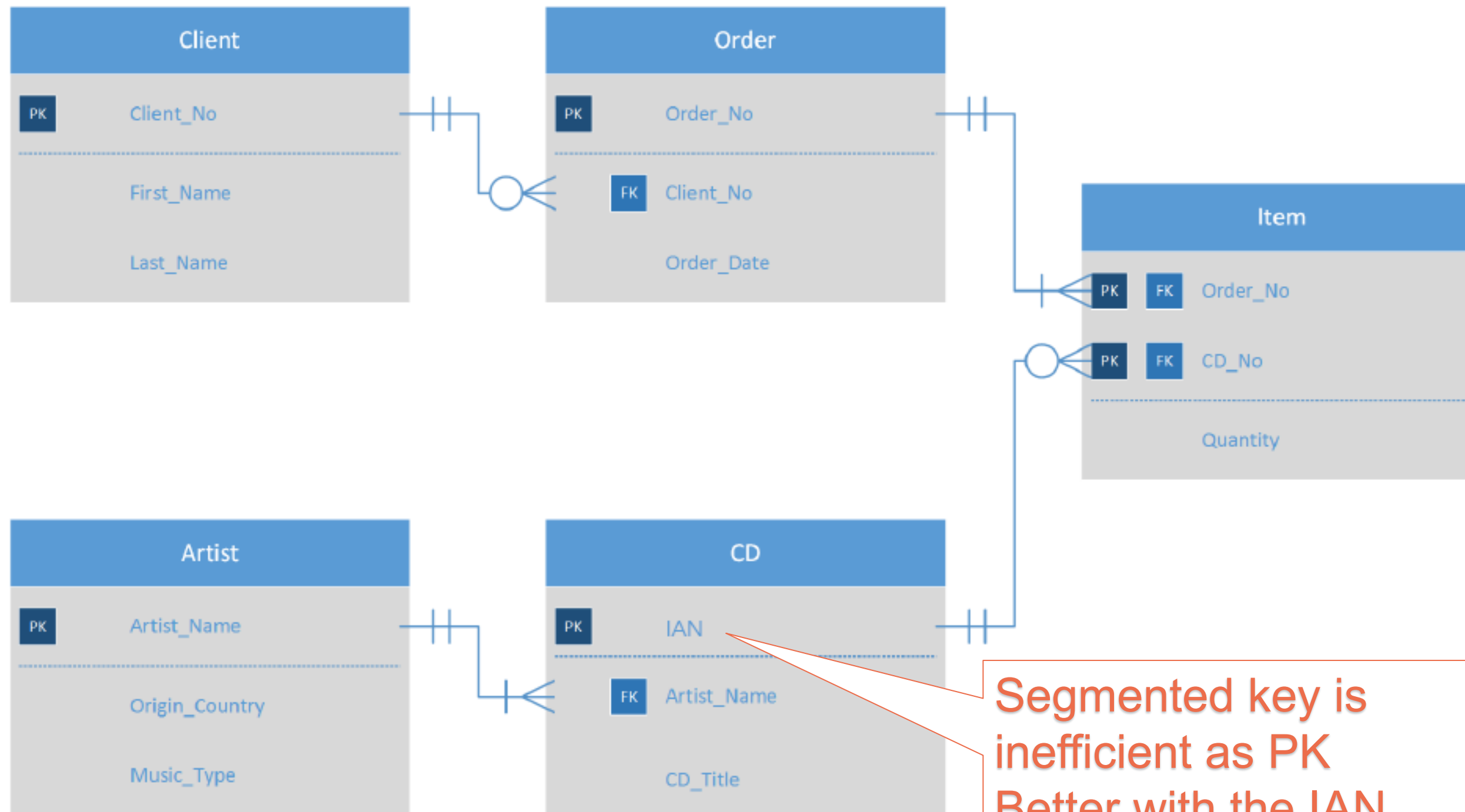


- Now create a Crow's Foot diagram as logical model with three exemplary attributes, including PK and – where needed – FK

Music shop – Martin Notation

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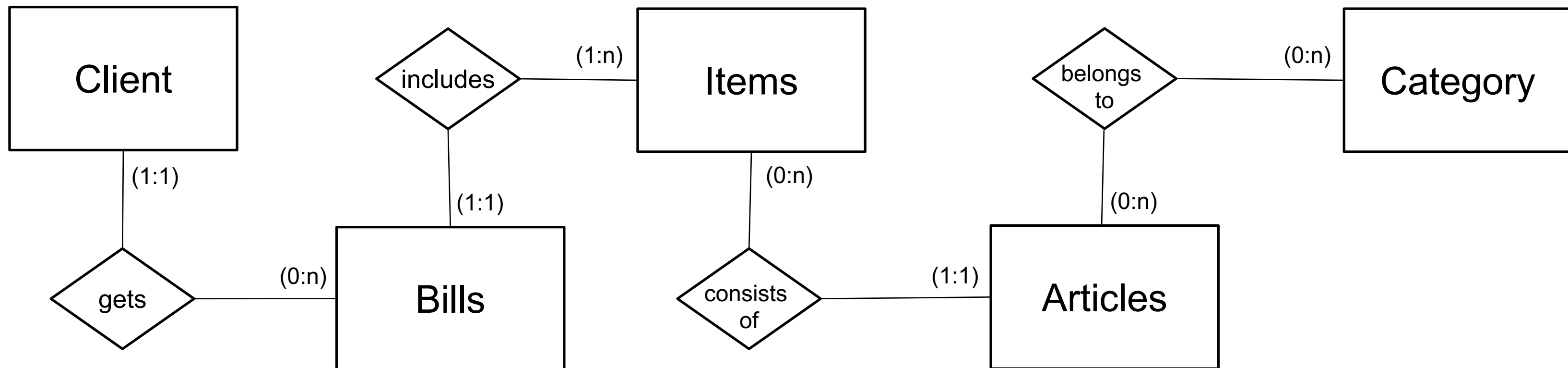


- The Company Wine & Vinegar ...
 - sells Wine, Vinegar, and Oil on tap
 - needs a database for the sales

- The client (Wine & Vinegar) demands:
 - Overview of earnings
 - Sums – month, quarter, year
 - Number of sold articles
 - Basis for mailing to the clients and potential clients (just the addresses – the mailings are produced outside)
 - No inventory control

- Open questions (and answers) - Examples
 - Are there any bundles and volume deals
 - ✧ No – everything is sold in milliliters
 - Are there special offers or regularly changing prices?
 - ✧ No, but sometimes we change the price individually for an item
 - Are there variations in one article type (e.g. year, class)
 - ✧ No
 - Will the articles be clustered in categories (e.g. type, quality, organic).
Has every article only one category?
 - ✧ Yes, there are categories. The articles can have multiple categories
 - Are there clustered bills? How is the payment procedure?
 - ✧ No, one sale with some articles at a time. Payment at once

- Chen Diagram (without attributes)

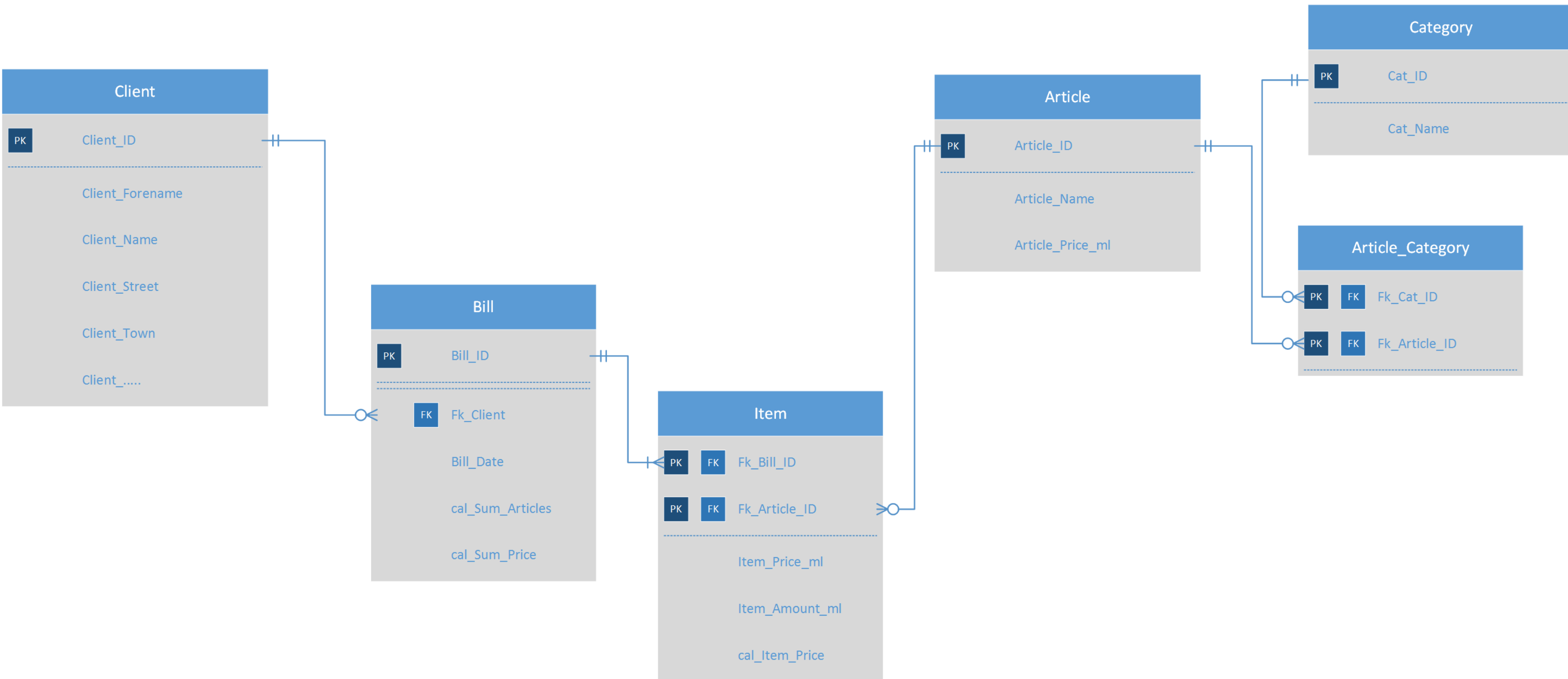


- There is no entity needed for the earnings – they are just a view created from the data in bills/items
- The list for the mailings is created as a view from client

Wine & Vinegar (4)

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Department

Normalize this form – Create Martin Notation

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Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Supervisor	Department	Department_Mgr	Department_Location
John	B	Smith	123456789	01.09.65	731 Fondren, Houston, TX	M	30000	Franklin T Wong, 333445555	Research	Franklin T Wong, 333445555	Bellaire, Sugarland, Houston
Franklin	T	Wong	333445555	08.12.55	638 Voss, Houston, TX	M	40000	James E Borg, 888665555	Research	Franklin T Wong, 333445555	Bellaire, Sugarland, Houston
Jennifer	J	Zelaya	999887777	19.01.68	3321 Castle, Spring, TX	F	25000	Alicia S Wallace, 987654321	Administration	Alicia S Wallace, 987654321	Stafford
Alicia	S	Wallace	987654321	20.06.41	291 Berry, Bellaire, Tx	F	43000	James E Borg, 888665555	Administration	Alicia S Wallace, 987654321	Stafford
Ramesh	K	Narayan	666884444	15.09.62	975 Fire Oak, Humble, Tx	M	38000	Franklin T Wong, 333445555	Research	Franklin T Wong, 333445555	Bellaire, Sugarland, Houston
Joyce	A	English	453453453	31.07.72	5631 Rice, Houston, TX	F	25000	Franklin T Wong, 333445555	Research	Franklin T Wong, 333445555	Bellaire, Sugarland, Houston
Ahmad	V	Jabbar	987987987	29.03.69	980 Dallas, Houston, TX	M	25000	Alicia S Wallace, 987654321	Administration	Alicia S Wallace, 987654321	Stafford
James	E	Borg	888665555	10.11.37	450 Stone, Houston, TX	M	55000	NULL	Headquarters	James E Borg, 888665555	Houston

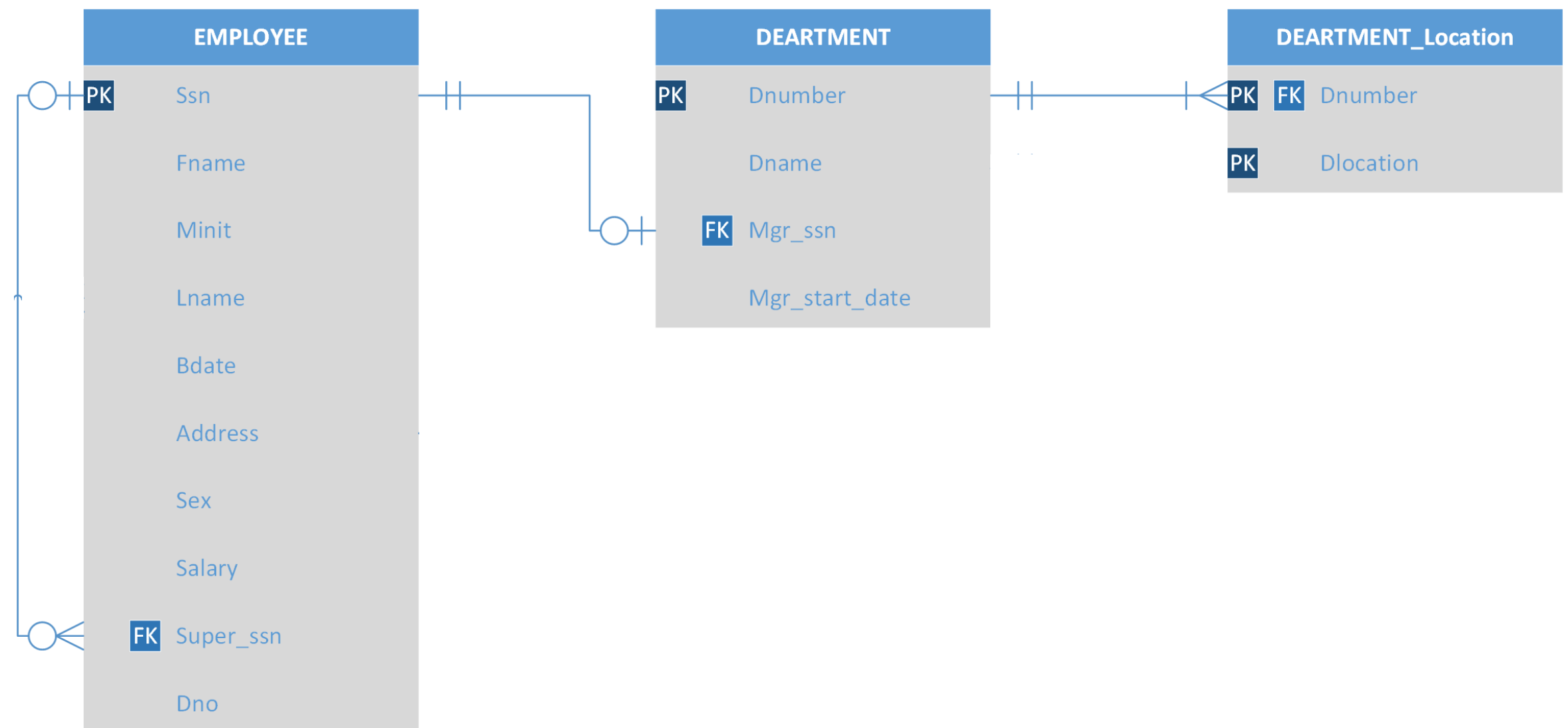
Keep
Address as
one Element

Department Martin Notation

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■ Solution

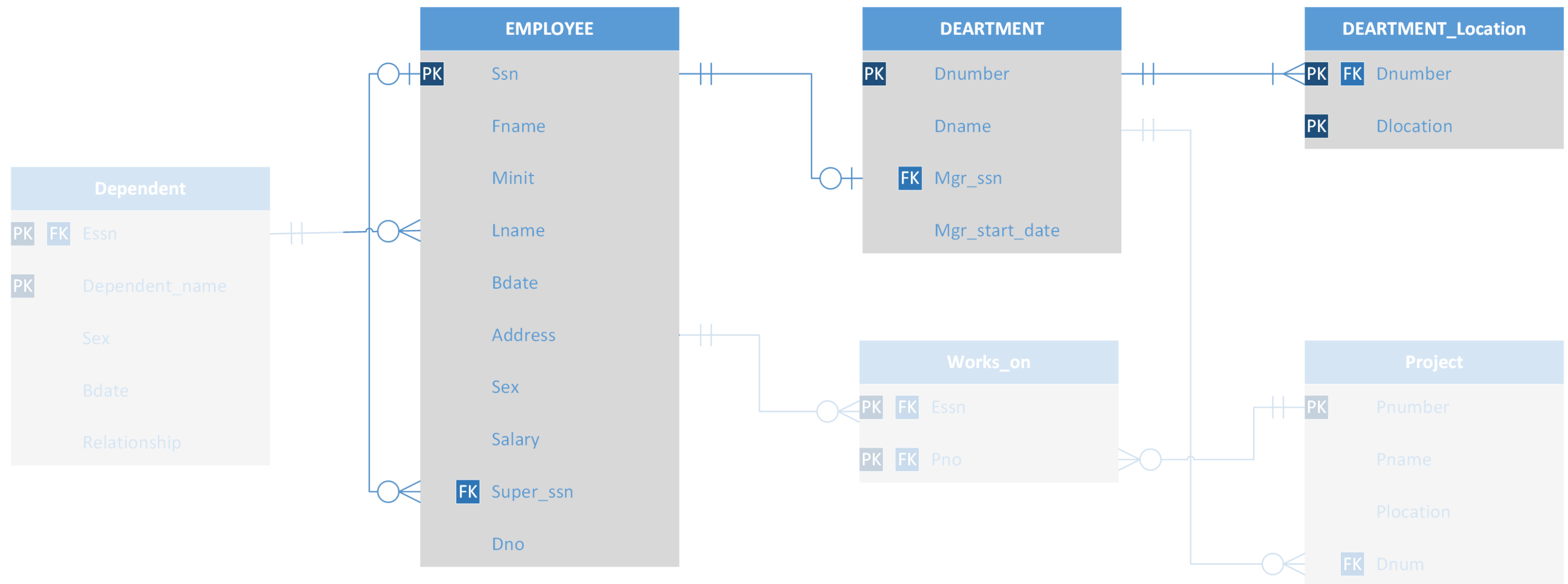


Department Martin Notation

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- Solution (the lighter ones indicate a more complete version)



More examples – for self studies

- Create the database of an online dating platform
 - Each member can send messages on each other member
 - Each message has one sender, one recipient, a title, a body, and a timestamp
 - Extend it – what happens, when there could be multiple recipients?
 - Each member can block each other member from sending messages
- A travel agency gathers the data of travels in a database
 - Each client books one or many trips
 - Each employee can sell trips to all clients
 - Each booked trip has one payment type
 - Each client has one contact person in the agency (but may buy from another employee)
- An IT-seminar company gets a new database.
 - Each lecturer gives at least one seminar.
 - Each type of seminar is offered multiple times
 - A seminar is given in one day (Just store the date)
 - Highly demanded seminars are given by different lecturers at the same time
 - Each client can book one or more seminars
 - We do not check if there are time schedule clashes when multiple seminars are booked
 - There are no prerequisite seminars for a seminar
 - Each seminar timeslot takes only place when it is booked from min 5 clients – the maximal number varies depending on the seminar.